THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 20	Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.
2	**(a)** -14	Put $p(z) = 0$.
3	**(b)** $\frac{39}{65}$	Reduce the fraction; only 2 and 5 may remain in the denominator.
4	**(a)** $49x^2 + 56xy + 16y^2$	Use the square identity.
5	**(b)** $(4.5,\ 4.5)$	Average the coordinates.
6	**(b)** 3	Slope $= \frac{y_2 - y_1}{x_2 - x_1}$.
7	**(d)** 4	$400 = 2^4 \times 5^2$
8	**(a)** $6x^2 + 19x + 15$	Multiply term by term.
9	**(b)** $(-1, -3)$	The centre is the midpoint of a diameter.
10	**(c)** -3	Put $y = 0$.
11	**(b)** $\frac{4}{33}$	$x = \frac{ab}{99}$, then reduce.
12	**(b)** 45	$a^2 + b^2 = (a - b)^2 + 2ab$.
13	**(a)** $(3, 0)$	Set $PA = PB$ with $P = (x, 0)$.
14	**(b)** 4	Substitute y and solve for x.
15	**(a)** 6	Find how many digits repeat.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$d^2 = (4 - (1))^2 + (5 - (1))^2 = 25$ (1); $d = 5$ units (1).	1+1
17	$m = \frac{13 - (4)}{5 - (2)}$ (1); $m = 3$ (1).	1+1
18	Multiply by 10 and by 100 and subtract: $90x = 75$ (1); $x = \frac{5}{6}$ (1).	1+1
19	Multiply term by term (1); $= 6x^2 + 17x + 5$ (1).	1+1
20	$\left(\frac{3+(6)}{2}, \frac{-4+(7)}{2}\right)$ (1); Midpoint $= (4.5,\ 1.5)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$AB = 1 \times 5 = 5$ (direction $(3, 4)$); $BC = 2 \times 5 = 10$ (1); $AC = 15 = AB + BC$ (1); So the points are collinear (1).	1+1+1
22	$18 = m(2) + 12$ (1); $m = 3$ (1); x-intercept: $0 = 3x + 12$, $x = -4$ (1).	1+1+1
23	$1000x - x = 27$ (1); $999x = 27$ (1); $x = \frac{1}{37}$ (1).	1+1+1
24	$x^2 + \frac{1}{x^2} = (6)^2 - 2 = 34$ (1); $\left(x - \frac{1}{x}\right)^2 = 34 - 2 = 32$ (2).	1+2
	OR: $(2 - k)^2 + 16 = 25$ (1); $(2 - k)^2 = 9$ (1); $k = -1$ or $k = 5$ (1).	1+1+1
25	Points $(0, 5)$, $(1, 1)$, $(2, -3)$ (1); Slope $= \frac{-3 - (5)}{2 - 0}$ (1); $= -4$ (1).	1+1+1
	OR: $x = 1 + 0.2\overline{3}$ (1); $0.2\overline{3} = \frac{7}{30}$ (1); $x = \frac{37}{30}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$AB^2 = 9 + 16 = 25$ (1); $BC^2 = DA^2 = CD^2 = 25$: all sides equal 5 (1); $AC = 2 \times 3 = 6$ (1); $BD = 2 \times 4 = 8$ (1); $6 \neq 8$, so not a square (1).	1+1+1+1+1
	OR: Slope $= \frac{16 - (8)}{3 - (1)} = 4$ (1); $p(x) = 4x + 4$ (1); Zero: $x = -1$ (1); $p(5) = 24$ (2).	1+1+1+2
27	Let $x = 1.1\overline{6}$; $10x$ and $100x$ give $90x = 105$ (1); $x = \frac{7}{6}$ (1); Long division: $1.1\overline{6}$ (1); Matches the given decimal (2).	1+1+1+2
	OR: Use $(a + b + c)^2$ with $a = 2x$, $b = -3y$ (1); Squares: $4x^2 + 9y^2 + 1z^2$ (1); Cross terms: $-12xy - 6yz + 4zx$ (1); $= 4x^2 + 9y^2 + z^2 - 12xy - 6yz + 4zx$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 7	Square both sides.
2	**(b)** Quadrant III	Use the sign conditions on the coordinates.
3	**(b)** $8x - 5$	Add like terms.
4	**(d)** 6	Multiply under the root.
5	**(c)** 1006009	Write $1003 = 1000 + 3$
6	**(c)** 36	Add the three side lengths.
7	**(b)** 4	$p(2) - p(1) = a$.
8	**(b)** rational	Collect the rational and irrational parts.
9	**(c)** 5600	Use $a^2 - b^2 = (a+b)(a-b)$.
10	**(a)** $(-4, -9)$	Both signs change.
11	**(a)** 82	Substitute n.
12	**(d)** $\frac{3}{7}$	Take the average.
13	**(d)** $(x-7)(x+4)$	Find the pair of numbers.
14	**(c)** 15	Area = length × breadth.
15	**(d)** parallel	Use the slope-intercept form.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$p - q = 3x + 9$ (1); Zero $x = -3$ (1).	1+1
17	Long division of 5 by 22 (1); $= 0.2\overline{27}$, block 27 (1).	1+1
18	$x^2 + y^2 = (x+y)^2 - 2xy$ (1); $= 121 - 56 = 65$ (1).	1+1
19	Centre is the midpoint of the diameter (1); $B = (-3, -6)$ (1).	1+1
20	$a_1 = 5$, $a_2 = 13$, $a_3 = 21$ (1); $a_{10} = 77$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$a^2 + b^2 = 4^2 + 2(21)$ (1); $= 58$ (1); $(a+b)^2 = 16 + 4(21) = 100$ (1).	1+1+1
22	$AB^2 = 18$, $BC^2 = 18$; $CA^2 = 36$ (1); Two sides equal (1); $AB^2 + BC^2 = CA^2$: right angle at B (1).	1+1+1
23	$a_n = 7n - 1$ (1); $7n - 1 = 90$ (1); $n = 13$ (1).	1+1+1
24	Lowest terms: $\frac{7}{25}$ (1); Denominator $= 25 = 5^2$ (1); Terminates, since only 2 and 5 occur (1).	1+1+1
	OR: Split the middle term: $11x = 8x + 3x$ (1); Group the terms (1); $= (2x+3)(x+4)$ (1).	1+1+1
25	$AB = 2 \times 13 = 26$ (direction $(5, 12)$); $BC = 1 \times 13 = 13$ (1); $AC = 39 = AB + BC$ (1); So the points are collinear (1).	1+1+1
	OR: $32 = m(3) + 20$ (1); $m = 4$ (1); x-intercept: $0 = 4x + 20$, $x = -5$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$D = A + C - B = (2, 5)$ (1); Midpoint of $AC = (3.5, 3)$ (1); Midpoint of $BD = (3.5, 3)$ (same point) (1); Diagonals bisect each other (1); Hence $ABCD$ is a parallelogram (1).	1+1+1+1+1
	OR: $E(n) = 150n + 5000$ (1); $E(20) = 8000$ (1); $150n + 5000 = 14000$, $n = 60$ (1); Extra items $= \frac{1200}{150} = 8$ (2).	1+1+1+2
27	$(3 + \sqrt{5})^2 = 14 + 6\sqrt{5}$ (1); $(3 - \sqrt{5})^2 = 14 - 6\sqrt{5}$ (1); Sum $= 28$ (1); The surd terms cancel (1); Result is rational (1).	1+1+1+1+1

	OR: $(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ca)$ (1); $81 = a^2+b^2+c^2+52$ (1); $a^2+b^2+c^2 = 29$ (1); Substitution checked (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 11	Apply Brahmagupta's rules in order.
2	**(a)** 32	$4xy = (x+y)^2 - (x-y)^2$.
3	**(c)** 65	Use $d^2 = (x_2-x_1)^2 + (y_2-y_1)^2$.
4	**(a)** -30	Put $p(z) = 0$.
5	**(c)** $\frac{21}{70}$	Reduce the fraction; only 2 and 5 may remain in the denominator.
6	**(d)** $36x^2 - 60xy + 25y^2$	Use the square identity.
7	**(c)** $(-1.5,\ 6.5)$	Average the coordinates.
8	**(c)** 3	Slope $= \frac{y_2-y_1}{x_2-x_1}$.
9	**(a)** 3	$250 = 2^1 \times 5^3$
10	**(d)** $6x^2 + 25x + 14$	Multiply term by term.
11	**(a)** $(3, 0)$	The centre is the midpoint of a diameter.
12	**(b)** 2	Put $y = 0$.
13	**(c)** $\frac{6}{11}$	$x = \frac{ab}{99}$, then reduce.
14	**(c)** 73	$a^2 + b^2 = (a-b)^2 + 2ab$.
15	**(d)** $(-2, 0)$	Set $PA = PB$ with $P = (x, 0)$.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$= 8^2 - (\sqrt{5})^2$ (1); $= 59$ (1).	1+1
17	$1002 = 1000 + 2$ (1); $1002^2 = 1004004$ (1).	1+1
18	$d^2 = (2-(-2))^2 + (6-(3))^2 = 25$ (1); $d = 5$ units (1).	1+1
19	$m = \frac{8-(0)}{3-(-1)}$ (1); $m = 2$ (1).	1+1
20	Multiply by 10 and by 100 and subtract: $90x = 55$ (1); $x = \frac{11}{18}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$1000x - x = 135$ (1); $999x = 135$ (1); $x = \frac{5}{37}$ (1).	1+1+1
22	$x^2 + \frac{1}{x^2} = (8)^2 - 2 = 62$ (1); $\left(x - \frac{1}{x}\right)^2 = 62 - 2 = 60$ (2).	1+2
23	$(-3-k)^2 + 144 = 169$ (1); $(-3-k)^2 = 25$ (1); $k = -8$ or $k = 2$ (1).	1+1+1
24	Points $(0, -6)$, $(1, 3)$, $(2, 12)$ (1); Slope $= \frac{12-(-6)}{2-0}$ (1); $= 9$ (1).	1+1+1
	OR: $x = 2 + 0.4\overline{6}$ (1); $0.4\overline{6} = \frac{7}{15}$ (1); $x = \frac{37}{15}$ (1).	1+1+1
25	$a^2 + b^2 = 6^2 + 2(40)$ (1); $= 116$ (1); $(a+b)^2 = 36 + 4(40) = 196$ (1).	1+1+1
	OR: $AB^2 = 18$, $BC^2 = 18$; $CA^2 = 36$ (1); Two sides equal (1); $AB^2 + BC^2 = CA^2$: right angle at B (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$AB^2 = 25 + 144 = 169$ (1); $BC^2 = DA^2 = CD^2 = 169$: all sides equal 13 (1); $AC = 2 \times 5 = 10$ (1); $BD = 2 \times 12 = 24$ (1); $10 \neq 24$, so not a square (1).	1+1+1+1+1
	OR: Slope $= \frac{20-(5)}{6-(3)} = 5$ (1); $p(x) = 5x - 10$ (1); Zero: $x = 2$ (1); $p(4) = 10$ (2).	1+1+1+2
27	Let $x = 2.3\overline{3}$; $10x$ and $100x$ give $90x = 210$ (1); $x = \frac{7}{3}$ (1); Long division: $2.\overline{3}$ (1); Matches the given decimal (2).	1+1+1+2
	OR: Use $(a+b+c)^2$ with $a = 2x$, $b = -3y$ (1); Squares: $4x^2 + 9y^2 + 4z^2$ (1); Cross terms: $-12xy - 12yz + 8zx$ (1); $= 4x^2 + 9y^2 + 4z^2 - 12xy - 12yz + 8zx$ (2).	1+1+1+2

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